

# Krushank Bayyapu

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## Summary

Machine Learning Engineer with 3+ years building scalable generative AI platforms, agentic systems, and LLM infrastructure at enterprise scale. M.S. + B.S. in Computer Science with focus in AI and ML from the University of Illinois Urbana-Champaign

## Professional Experience

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Early Career Digital Leadership Program (2 Year Rotational Program)

August 2025 – Present

*United Airlines*

*Chicago, Illinois*

February 2026 – Present

### **Rotation 2: Agentic AI Systems Engineer**

- Designed and owned a production-grade agentic AI platform enabling autonomous workflows across internal operations, serving 150+ users and reducing manual review time by 73% (~300+ hours/month saved)
- Built and deployed a RAG-based knowledge system in production, processing 100K+ documents and reducing issue resolution time by 47% through optimized retrieval + prompt orchestration
- Developed LLM evaluation, monitoring, and feedback pipelines (automated ground truth + scoring), improving response quality by 25% and enabling continuous model governance
- Architected a multi-agent orchestration framework with tool use + planning, powering cross-functional workflows and increasing task completion rates by 2.5x
- Led integration of AI systems into internal platforms, driving 4+ point NPS improvement and adoption across multiple engineering and operations teams

### **Rotation 1: Machine Learning – LLM Engineer**

August 2025 – February 2026

- Designed and deployed core components of an enterprise LLM platform SDK, enabling 10+ teams to build and scale production AI applications across the organization
- Built multi-model routing + inference pipelines (OpenAI, Bedrock, Whisper, Veo), reducing inference cost by 32%+ while improving latency and reliability
- Implemented automated evaluation + ground truth generation system supporting large-scale LLM benchmarking and production monitoring
- Developed and deployed production-ready AI agents for code generation and workflow automation, improving developer productivity by 40%
- Led Atlassian MCP integration into production AI platform, enabling seamless agent interaction with enterprise tools and increasing automation coverage across workflows
- Scaled platform capabilities to support millions of LLM requests per month across internal use cases

Software Engineer Intern

May 2025 – July 2025

*TrustLogix.AI*

*Mountain View, California, United States*

- Built and deployed an AI-driven test automation framework leveraging LLM agents, reducing manual QA effort by 60% and increasing test coverage across core workflows
- Designed end-to-end testing pipelines integrated into CI/CD, improving release confidence and reducing regression bugs by 35%
- Migrated legacy test systems to structured, scalable formats, improving cross-team collaboration and reducing onboarding time for QA engineers
- Developed production-grade tooling used across engineering teams to maintain system reliability at scale

Cyber Digital Risk Technology Intern

May 2024 – August 2024

*United Airlines*

*Chicago, Illinois, United States*

- Designed and deployed AI-powered fraud case management workflows used in production for enterprise fraud detection and response
- Automated fraud resolution pipelines, reducing case handling time by 38% and improving customer resolution speed
- Integrated Redshift + Splunk data pipelines, enabling real-time monitoring and analytics across fraud systems
- Built interim SOAR-based automation system, improving incident response efficiency and operational scalability

GenAI Research Development Assistant, FORWARD Data Lab

August 2023 – December 2023

*University of Illinois Urbana-Champaign*

*Champaign, Illinois, United States*

- Designed and implemented a next-generation AI search system leveraging LLM planning + retrieval, supporting large-scale knowledge discovery across millions of documents
- Built end-to-end retrieval + reasoning pipeline, combining LLMs with structured data systems to improve answer relevance by more than 57%

- Prototyped system architectures that informed production-ready RAG system design patterns

GenAI Digital Technology Intern

May 2023 – August 2023

United Airlines

Chicago, Illinois, United States

- Built and deployed an LLM-powered retrieval system for enterprise knowledge access, improving internal search efficiency by 63%
- Fine-tuned and optimized models for extended context handling, improving response completeness and accuracy by 33%
- Developed reproducible pipelines for document ingestion, retrieval, and inference, later integrated into broader AI initiatives

Intern- Associate Data Engineer

May 2022 – August 2022

Blue.cloud

Chicago, Illinois, United States

- Built scalable data pipelines integrating APIs and cloud systems (Snowflake, Azure), supporting downstream ML workflows
- Enabled real-time data processing and analytics, improving insight generation speed across datasets

Education

University of Illinois Urbana-Champaign

Champaign, Illinois, United States

Master of Computer Science (MCS)

Graduated December 2025

- Highest Honors Graduate

University of Illinois Urbana-Champaign

Champaign, Illinois, United States

Bachelor of Science (BS), Computer Science

Graduated December 2024

- Dean’s List, Edmund J. James Scholar Honors

Skills

**Languages:** Python, Java, C, C++, R, Bash, Git, MATLAB  
**AI / ML:** Agentic AI, Large Language Models (LLMs), Generative AI, Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), Machine Learning, LangChain, LangGraph, Model Context Protocol (MCP)  
**Cloud & Data:** AWS, Azure Data Factory, Snowflake, Databricks, MySQL, MongoDB, Docker  
**Tools & Platforms:** Playwright, Kubernetes, Splunk SOAR, PEGA, Webhook APIs

Certifications & Accreditations

- AWS Certified AI Practitioner | AWS | 2026
- Leadership Institute - Credential II | United Airlines | 2026
- AI Agent Fundamentals | Databricks | 2026
- Model Context Protocol: Advanced Topics | Anthropic | 2025
- Introduction to Model Context Protocol | Anthropic | 2025
- Generative AI Fundamentals | Databricks | 2024
- Databricks Lakehouse Fundamentals | Databricks | 2024
- AWS Certified Cloud Practitioner | AWS | 2024
- Generative Adversarial Networks (GANs) Specialization | DeepLearning.AI | 2023

Technical Side Projects

AI Interactive Code Dependency Visualizer

September 2025

- Built local AI assistant using Llama 3, Langchain, and Neo4j to visualize and analyze code dependencies
- Extracted and summarized file dependencies with Python AST and Llama3
- Created interactive dependency graphs with Neo4j and Pyvis for high-level and detailed insights
- Developed an AI chat interface for contextual, file-aware Q&A across codebases
- Maintained privacy with a local-first design using Ollama and Streamlit for interactive visualizations

LLM-Powered Instructional Training System

May 2025

- Built a production-style multi-agent training platform enabling scenario simulation and adaptive, LLM-driven feedback loops
- Designed a persistent memory + retrieval system supporting multi-session learning and strong contextual continuity
- Leveraged Meta’s Llama3 with prompt engineering and RAG-based ingestion to ground outputs in instructional materials and generate culturally aware, pedagogically aligned feedback